From Eliciture to Internal Representation: Implicit Causality in Language Models

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Abstract

Do language models merely behave as if they infer implicit causes, or do they encode causal relationships that generalize? Four studies test behavioral elicitation, relation selectivity, syntactic transfer, and naturalistic transfer. Across six models, behavioral conclusions depend on elicitation method, whereas causal directions discriminate held-out causal pairs and reach .75–.94 accuracy after a cross-sentence to within-sentence shift. Demographic substitutions also shift causal projections in 65 matched scenarios per model. These studies connect pragmatic behavior to internal representation while exposing important construct-validity limits.

1 Introduction

Cohen and Kehler call defeasible pragmatic enrichments *conversational elicitures* [Cohen and Kehler, 2021]. Language models reproduce human-like eliciture judgments and relative-clause preferences [Pan and Kehler, 2025], but behavioral agreement does not reveal whether they encode the inferred relation. Because linear representations can reveal event-plausibility distinctions obscured by output probabilities [Lepori et al., 2026], we use contrastive activation addition (CAA) [Rimsky et al., 2024] to study implicit causality.

Study 1 tests whether behavioral evidence is robust to measurement; Study 2 compares causal, temporal, and plausibility directions; Study 3 tests syntactic transfer; and Study 4 tests matched naturalistic scenarios. Behavioral measures disagree, whereas causal directions are discriminative, relation-selective, and informative after both shifts.

2 Study 1: Measuring Behavioral Eliciture

We vary the response measure in a conceptual replication of Pan and Kehler, finding substantial measurement dependence.

2.1 Methods

Models. We evaluate the base checkpoints of Gemma 2 2B/9B [Gemma Team, 2024], Qwen3 4B/8B [Qwen Team, 2025], Llama 3.2 3B [Meta AI, 2024], and Llama 3 8B [Dubey et al., 2024].

Dataset and analysis. We evaluate 150 critical/control pairs with yes/no judgments, expected seven-point Likert scores, and the relative probability of the same continuation after *because* versus bare *and*. The last measure uses no metalinguistic prompt. A pair is correct when the critical item receives the stronger causal response, allowing us to compare prompting with probability measurement [Hu and Levy, 2023].

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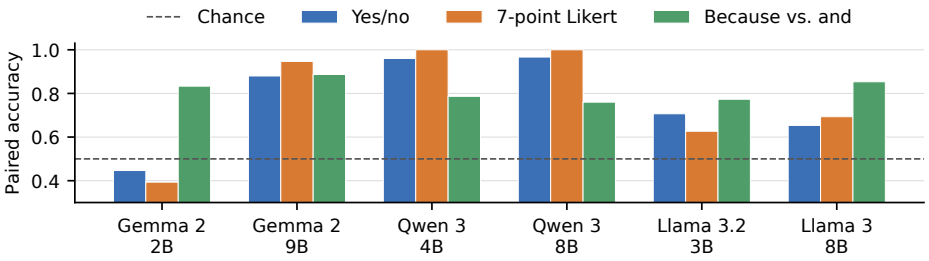


Figure 1: **RQ1.** Paired accuracy on 150 eliciture pairs. Dashed line: chance. Behavioral conclusions vary substantially with response mapping, particularly for smaller models.

2.2 Results

Figure 1 shows that Qwen3 4B/8B reach 1.00 with Likert responses and at least .96 with yes/no, but fall to .79/.76 under probability assignment. Conversely, Gemma 2 2B scores .45/.39 with prompting but .83 with *because* versus *and*. Overall, no interface supports a model-independent conclusion.

3 Study 2: Relation-Selective Causal Representations

We compare cross-sentence causal directions with temporal and plausibility directions, finding strong diagonals and weaker cross-transfer.

3.1 Methods

Datasets. Causal, temporal, and plausibility directions respectively use 384/96, 800/200, and 56/14 train/ID-development pairs.

Analysis. At every fourth layer and the final layer, CAA averages final-token differences between matched positive and negative texts. We select the earliest layer maximizing mean own-development paired accuracy and ROC–AUC, then apply that frozen direction to every target relation. Matrix diagonals measure within-relation discrimination; off-diagonals measure cross-transfer. This is an ID-development analysis, not an OOD test.

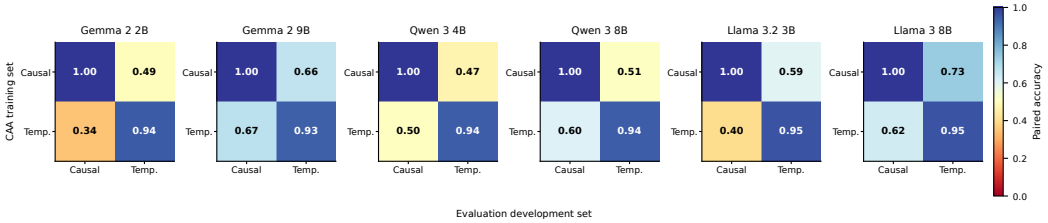


Figure 2: **RQ2a, causal vs. temporal.** ID-development paired accuracy when each row's CAA direction is evaluated on each column's relation. Each panel is one model; diagonals measure own-relation discrimination and off-diagonals measure cross-transfer.

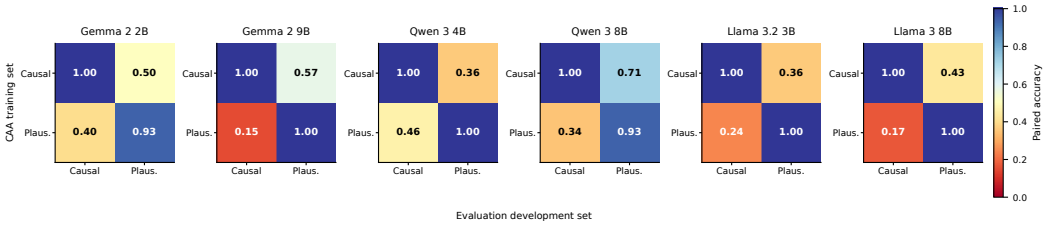


Figure 3: **RQ2a, causal vs. plausibility.** Six-model cross-transfer matrices; plausibility cells ($n = 14$ pairs) are exploratory.

3.2 Results

Figures 2–3 show causal self-transfer of 1.00 for all six models, temporal self-transfer of .93–.95, and plausibility self-transfer of .93–1.00. However, causal-to-temporal accuracy is .47–.73 and causal-to-plausibility is .36–.71; reverse transfer reaches .34–.67 and .15–.46. The gaps support relation selectivity, while occasional above-chance transfer argues against strict orthogonality.

4 Study 3: Generalization Across Syntax

We next test whether cross-sentence causal directions generalize to within-sentence eliciture.

4.1 Methods

We apply each Study 2 causal direction to 200 held-out within-sentence relative-clause pairs, changing both syntax and discourse unit [Rohde et al., 2011]. A pair is correct when the critical sentence projects farther along the direction.

Table 1: **RQ2b.** Cross-sentence causal CAA paired accuracy on 200 within-sentence pairs.

	Gemma 2 2B	Gemma 2 9B	Qwen 3 4B	Qwen 3 8B	Llama 3.2 3B	Llama 3 8B
Paired accuracy	.880	.935	.750	.785	.835	.805

4.2 Results

Table 1 shows .75–.94 paired accuracy; all six exceed chance. Cross-sentence causal coherence and within-sentence eliciture therefore share a linearly accessible component, although it need not be uniquely causal.

5 Study 4: Generalization to Naturalistic Scenarios

We finally test causal representations on matched naturalistic scenarios.

5.1 Methods

Dataset. We evaluate 65 self-blinding scenarios per model, with matched White-male, minority-male, and minority-female versions.

Analysis. A unit causal direction fit separately on 1,000 within-sentence pairs projects each scenario's final-token state. We report within-scenario differences from the White-male version with 10,000-sample bootstrap 95% intervals.

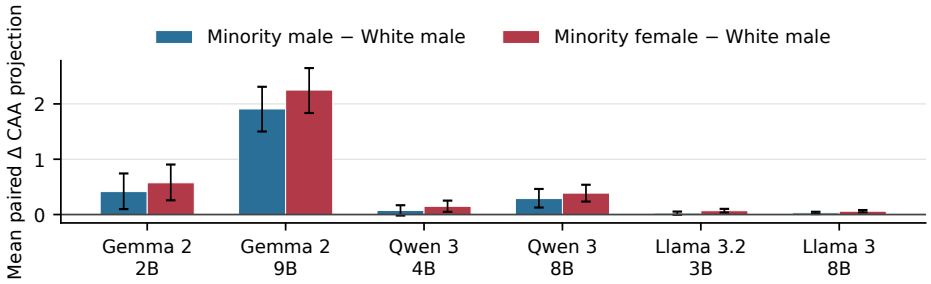


Figure 4: **RQ3.** Mean paired CAA-projection change (65 scenarios/model); bootstrap 95% intervals.

5.2 Results

Figure 4 shows positive point estimates for both contrasts in every model. Minority-female intervals exclude zero for all six; minority-male intervals do so for four. Notably, minority-female magnitudes

range from .06–.07 in Llama models to 2.25 in Gemma 2 9B. These associations do not establish social bias, downstream behavior, or a causal mechanism.

6 Discussion and limitations

Across four studies, behavioral evidence is measurement-sensitive, whereas causal directions are relation-selective and transferable. However, Study 2 uses ID development sets (only 14 plausibility pairs), CAA is correlational, and 65 scenarios cannot establish general social behavior. OOD tests, causal interventions, and behavioral validation remain necessary.

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